LocalCloudManager

Interface Design Description

Eclipse Arrowhead 5.2.2 | support system | Revision 2026-11-09 | REVIEW DRAFT

Purpose and document structure

This document consolidates the system interface designs. Detailed Java service/transport IDDs are retained as separate appendices, followed by available Go and Rust interface descriptions. The current cross-language sections below precede the detailed implementation appendices. Appendix dates and release labels belong to the retained baseline, not to a newly qualified release. Existing descriptions have been assembled from the primary trees; no live interoperability or semantic-parity claim is made.

Java implementation

Primary folder: Eclipse-Arrowhead-java/ah5-LocalCloudManager-java-spring. 2 detailed Java IDD source document(s) are included in the appendices. Their architecture, operation parameters, models, transport bindings, access rules and errors remain associated with Java. They must not be read as automatically implemented by the ports.

Go implementation

Primary folder: Eclipse-Arrowhead-go/ah5-LocalCloudManager-go.

LocalCloudManager - Go 5.2.2

Standalone support system, reorganized on 2026-10-28 following the Eclipse-Arrowhead-java distribution layout. Build and operate from this folder; no original Arrowhead checkout or sibling source module is required. The executable is bin/LocalCloudManager. Entry point: ./cmd/localcloudmanager. Module: arrowhead/ah5-support-go/localcloudmanager. See DEPLOYMENT.md for configuration and standalone container use. Existing compatibility limits remain in the copied upstream documentation and UPSTREAM-README.md. Existing role selection and API behavior are retained. Documentation and license notices are included in this folder. Source provenance and import-only rewrites are listed in the distribution's SOURCE-MANIFEST.json. No running cloud or upstream source was changed. See [Graphical orchestration and security editor](GRAPHICAL-ORCHESTRATION.md) for local rule and policy editing.

Listening port

Default 8480 on all interfaces. Override with SERVER_PORT=N (or PORT=N) or --server.port N. See [LISTENING-PORTS.md](../LISTENING-PORTS.md) for precedence and all systems.

Detailed implementation evidence

3 native Go IDD document(s) are included in the appendices. Their protocol and storage descriptions apply to that implementation only. Current primary-folder ownership and current listening-port notes above supersede historical paths or example ports in those retained documents.

Rust implementation

Primary folder: Eclipse-Arrowhead-rust/ah5-LocalCloudManager-rust.

Rust LocalCloudManager

Current implementation - secure Rust candidate cloud (2026-10-17)

The LocalCloudManager executable now uses src/secure_main.rs and src/secure.rs, not the historical HTTP launcher/router described below. It requires --config CONFIG.json. Automatic legacy binary discovery is disabled. The manager depends on the parent Rust support library for verified TLS, expected-provider CN/profile pinning, certificate-bound identity login/token refresh and bounded HTTPS requests. Deployed console: https://127.0.0.1:10590/. System API: https://127.0.0.1:10520/localcloud/ (mandatory mTLS). Deployment: Arrowhead-test-cloud-rust/candidate-native/, project arrowhead-rust-candidate. The older Rust cloud remains separate. During final verification it was recreated by a separate deployment change and its UI remapped from8590to10490; this work did not modify that project. The browser console uses HTTPS plus a private-password login and Secure/HttpOnly/SameSite cookie. It does not require a browser client certificate. The candidate CA is private, so browsers may show a certificate warning; no global trust store was changed. Password is in candidate-native/secrets/localcloudmanager/console-password (private file; never logged). The manager's system listener requires a verified local sy/op certificate and a matching verified sysop identity token for management. Echo is certificate-only. See docs/SECURE-DEPLOYMENT.md for the current configuration, service contracts, permission boundaries, test evidence and limitations. The manager now makes real revision-checked SysMonitor subscription writes, forwards plant POST/PUT/PATCH/DELETE correctly, manages simple-store rules, reads diagnostics/historian data and uses a scoped Rust Docker gateway rather than receiving a Docker socket. The Markdown below, the relocation source manifest, the earlier generated LaTeX/PDF catalogue and docs/legacy/ record the superseded relocation snapshot, not the active deployed secure contract. They do not supersede the new secure deployment documentation. Full Java equivalence, telemetry broker publishing and the cloud's missing federation/onboarding capabilities are still not claimed.

Historical relocation record (superseded runtime)

Canonical relocated source: ah5-support-rust/LocalCloudManager. Target system version5.2.2; associated services localCloudManagement and legacy monitor, both1.0.0. The latest located cloud/workspace manager sources matched SHA256 c3150daa4dd91e854439e31626895c5ff86e77881c338c5281306bda49f44d6b before relocation. This is a source/documentation relocation, not a Java-parity upgrade or secure deployment. The existing Arrowhead-test-cloud-rust/core-systems copy remains a deployment compatibility snapshot because other roles share its runtime and build context. No live source, cloud, container, key or state was removed or restarted. Develop against this canonical package instead of editing both copies independently.

Critical boundary

The inherited manager is a development HTTP adapter: no native inbound TLS, no configured mTLS provider identity and no native AMQP sysMonitor publisher. It sends/accepts legacy Bearer SYSTEM//... identities in its shared runtime. Those are not cryptographic authentication. Do not expose this package directly or assume it meets the newer Rust support runtime's certificate-authenticated contract. A TLS proxy alone does not correct legacy provider authentication. See COMPATIBILITY.md before use. The dedicated launcher defaults to http://127.0.0.1:8480/ instead of the inherited all-interface bind. This is a packaging safeguard, not a complete security fix. Historical UI8590/API8580 mappings belong to cloud deployment forwarding and are not created here.

Build and source layout

This is an independent nested Cargo package, with its own Cargo.toml/lock and workspace boundary. Root Makefile build/test/check targets explicitly include it. Root make localcloudmanager builds it; make localcloudmanager-docs generates/compiles its documentation. The generic parent arrowhead-support launcher does not select this legacy manager profile. src/main.rs is the dedicated launcher, rejecting roles other than localcloudmanager without mutating process environment. src/localcloudmanager.rs retains the existing Axum console handlers and compile-time embedded HTML/CSS/JavaScript in src/ui. api.rs, registry.rs and state.rs are

the inherited shared runtime dependencies; retaining them avoids a dependency on the old checkout. Other role implementations remain source dependencies and are not selected by this executable. No Java/Go/Node process implements the manager. Original manager business logic and UI are retained; deliberate packaging changes are the crate name/workspace boundary, library exports, role-default configuration helper, dedicated launcher and loopback default. Additional tests, documentation and source-provenance records are provided. The legacy src/ui.html remains an unused historical asset; active UI uses src/ui/index.html/app.js/styles.css.

Configuration

Environment	Actual source behavior
ARROWHEAD_ROLE	Omit or use localcloudmanager; other values rejected by dedicated launcher
SYSTEM_NAME, DOMAIN_NAME	Default LocalCloudManager/localcloudmanager; used as declared identity/advertised address, not validated certificate identities
SERVER_PORT, LISTEN_ADDRESS	Default8480 and loopback; explicit listen address can override it, but exposes inherited insecure transport
DATA_FILE	Independent JSON state, default /data/state.json; use a private dedicated path, never existing active cloud state
SERVICE_REGISTRY_URL, AUTHENTICATION_URL	Inherited HTTP defaults; HTTPS URLs alone do not provision a client certificate, expected-CN pin or identity-token login
DATAMANAGER_URL, SIMPLE_ORCHESTRATION_URL, SYSMONITOR_URL, PLANT_DESCRIPTION_ENGINE_URL	Provider adapter bases; shapes must be qualified against actual deployed providers
MANAGEMENT_WHITELIST, TOKEN_SECRET	Shared legacy runtime settings, not the console session boundary or proof of secure identity
CONSOLE_PASSWORD_HASH	scrypt:base64salt:base64hash, N16384/r8/p1; salt>=16bytes/hash>=32bytes; missing or invalid value returns503for console APIs
CONSOLE_ALLOWED_HOSTS	Host-name allowlist, default127.0.0.1,localhost,[::1]; ports are ignored
CONSOLE_SESSION_SECONDS	Default28800; positive bounded values should be provisioned; inherited implementation lacks robust range validation
DOCKER_SOCKET, COMPOSE_PROJECT_NAME	Unix Docker socket and project filter; static source allowlist controls lifecycle; MANAGED_SERVICES is not used

Missing password fails closed for /api/ but does not disable static assets, health or legacy Arrowhead routes. Sessions and history reset on restart. Keep this process in an isolated development environment with explicit trust boundaries.

Documentation and provenance

../documentation/LocalCloudManager/v5.2.2/ contains SysD, SysDD, two SDs and two HTTP IDDs, plus a combined catalogue. No sysMonitor AMQP SD/IDD is fabricated for an unimplemented publisher. The revised SysML variability model in docs/LocalCloudManager.sysml records Java-cloud, Go and Rust variants and the actual Rust qualification boundary. Historical Markdown/SysML is preserved unchanged under docs/legacy, not endorsed as evidence of interchangeability. Run from this module. The original source defaults to ../../Arrowhead-test-cloud-rust/core-systems for read-only source hashing; override REFERENCE when needed. docs/source-manifest.json records original and relocated hashes plus the route inventory. Read

VERIFICATION.md for actual test/build evidence. The repository's original Arrowhead LaTeX class/logo is reused.

Listening port

System API default 127.0.0.1:8480. Override with "listen" in CONFIG.json, SERVER_PORT=N or --server.port N. The HTTPS console listener ("consoleListen") is configured separately. See [LISTENING-PORTS.md](../LISTENING-PORTS.md) for precedence and all systems.

Detailed implementation evidence

2 native Rust IDD document(s) are included in the appendices. Their protocol and storage descriptions apply to that implementation only. Current primary-folder ownership and current listening-port notes above supersede historical paths or example ports in those retained documents.

Build and state ownership

Core entry points are built by arrowhead-core-rust; the ordinary support entry points are built by arrowhead-support-rust using the shared support runtime. LocalCloudManager has its own crate. Per-system documentation links may point into the shared documentation catalog. Independent identity and state configuration is still required. Read the native appendix for persistence and restart semantics.

Cross-language qualification

Identical system names or HTTP paths do not establish interoperability. For the selected implementation, verify registered service versions and metadata, method/path and payload models, null and optional-field handling, success/error bodies, authentication and per-operation authorization, timeout/retry behavior and persistence across restart. Java MQTT/AMQP descriptions apply to a port only where its own implementation evidence explicitly supports them. Missing transports must not be advertised. State conversion from Java SQL to native snapshots, secret provisioning, backup/restore and rollback require implementation-specific procedures. No data migration, deployment or cloud reconfiguration was performed for this document set.

Detailed appendices and provenance

- Java: Eclipse-Arrowhead-java/ah5-LocalCloudManager-java-spring/documentation/v5.2.2/LocalCloudManager-v5.2.2-localCloudManagement-HTTP-IDD.tex

- Java: Eclipse-Arrowhead-java/ah5-LocalCloudManager-java-spring/documentation/v5.2.2/LocalCloudManager-v5.2.2-sysMonitor-AMQP-IDD.tex

- Go: Eclipse-Arrowhead-go/ah5-LocalCloudManager-go/documentation/LocalCloudManager/v5.2.2/LocalCloudManager-Go-v5.2.2-localCloudManagement-HTTPS-IDD.tex

- Go: Eclipse-Arrowhead-go/ah5-LocalCloudManager-go/documentation/LocalCloudManager/v5.2.2/LocalCloudManager-Go-v5.2.2-monitor-HTTPS-IDD.tex

- Go: Eclipse-Arrowhead-go/ah5-LocalCloudManager-go/documentation/LocalCloudManager/v5.2.2/LocalCloudManager-Go-v5.2.2-sysMonitor-AMQP-IDD.tex

- Rust: Eclipse-Arrowhead-rust/ah5-support-runtime-rust/documentation/LocalCloudManager/v5.2.2/LocalCloudManager-Rust-v5.2.2-localCloudManagement-HTTP-IDD.tex

- Rust: Eclipse-Arrowhead-rust/ah5-support-runtime-rust/documentation/LocalCloudManager/v5.2.2/LocalCloudManager-Rust-v5.2.2-monitor-HTTP-IDD.tex

The package sources/ folder retains the selected LaTeX, figures, bibliography, language README/build evidence and source snapshots for newly described contracts. source-manifest.json records SHA-256 hashes of original inputs. build-report.json records whether an appendix used an existing newer PDF or was compiled from source.

localCloudManagement HTTP/TLS/JSON

Interface Design Description

Service ID: *"localCloudManagement"*

Abstract
This document describes the HTTP/1.1 protocol with TLS payload security and JSON payload encoding variant of the `localCloudManagement` service of the LocalCloudManager system.

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1 Overview

This document describes the `localCloudManagement` service interface of the LocalCloudManager system, which provides the operator console API of the local cloud: overview, registry and data views, orchestration rules, consumers, inter-cloud state, sysMonitor history and subscriptions, plant descriptions, logs and container lifecycle.

It's implemented using protocol, encoding as stated in below table:

Profile type	Type	Version
Transfer protocol	HTTP	1.1 [1]
Data encryption	TLS	1.3 [2], 1.2 [3]
Encoding	JSON	RFC 8259 [4]
Compression	N/A	-
Semantics	N/A	-
Ontology	N/A	-

Table 1: Communication and semantics details used for the `localCloudManagement` service interface

This document provides the Interface Design Description IDD to the *localCloudManagement – Service Description* document [5]. For further details about how this service is meant to be used, please consult that document.

The rest of this document describes how to realize the `localCloudManagement` service interface in detail. Both in terms of its operations (Section 2) and its information model (Section 3).

1.1 Interface registration

LocalCloudManager listens on two connectors. The Arrowhead connector (port 8491, HTTPS with mutual TLS) is registered in the ServiceRegistry with the interface template `generic_https` and policy `CERT_AUTH`, base path `/api`. The console connector (port 8490, plain HTTP, published on the host's loopback address only) serves the browser console with the same routes plus the console-only ones; it is not registered. The `operations` property lists the operations below with their method and path relative to the base path (path values are appended to the registered path):

Operation	Method	Path
`overview`	GET	`/overview`
`serviceRegistry Management`	GET	`/service-registry-management`
`dataManager`	GET	`/datamanager`
`consumers`	GET	`/consumers`
`interCloud`	GET	`/intercloud`
`sysMonitorHistory`	GET	`/sysmonitor`
`sysMonitorSubscriptions`	GET	`/sysmonitor/subscriptions`
`subscribeSysMonitor`	PUT	`/sysmonitor/subscriptions`
`unsubscribeSysMonitor`	DELETE	`/sysmonitor/subscriptions`
`plantDescriptions`	GET	`/plantdescriptions`
`createPlantDescription`	POST	`/plantdescriptions`
`replacePlantDescription`	PUT	`/plantdescriptions`
`updatePlantDescription`	PATCH	`/plantdescriptions`

Operation	Method	Path
`removePlantDescription`	DELETE	`/plantdescriptions`
`orchestrationRules`	GET	`/orchestration/rules`
`createOrchestrationRule`	POST	`/orchestration/rules`
`removeOrchestrationRule`	DELETE	`/orchestration/rules`
`logs`	GET	`/logs`
`lifecycle`	POST	`/containers`

`generalManagement` is registered separately (base path `/localcloudmanager/general/mgmt`, operation `get-config`, `GET /get-config?keys=...`, Sysop only).

1.2 Access

Arrowhead connector (8491). A request needs the caller's client certificate and `Authorization: Bearer IDENTITY-TOKEN//<token>` of the same system (common outsourced authentication filter; 401 otherwise, with the common error body `{"errorMessage", "errorCode", "exceptionType", "origin"}`). A registered operation is allowed for Sysop, the systems of `lcm.api.whitelist` and systems with a Consumer-Authorization grant of the operation (403 otherwise); other `/api` paths only for Sysop and the whitelist. The console-only paths (`/api/session`, `/health`, the UI) are 404 there.

```
GET /api/overview HTTP/1.1
Host: cloud-monitor-api:8491
Authorization: Bearer IDENTITY-TOKEN//<identity token>
```

Listing 1: An invocation on the Arrowhead connector.

Console connector (8490). `Host` and `Origin` must be allowed console names (403 `"Host or Origin is not an allowed console address"`; Origin `null` is refused); with a configured proxy secret, `/api` requests must carry `X-Console-Proxy` (403 `"Not forwarded by the console"`); without a configured password every `/api` request is 503. Every `/api` route except `/api/session` needs the session cookie `ah_console` (401 `"Console login required"`). Other paths serve the static console (`index.html` for unknown paths). Responses carry `X-Frame-Options: DENY`, `Content-Security-Policy: frame-ancestors 'none'`, `X-Content-Type-Options: nosniff` and `Referrer-Policy: no-referrer`.

2 Service Operations

The interfaces of the `localCloudManagement` service, its operations, data models and implementation are provided below. In particular, each subsection first names the HTTP method and path used to call the operation, after which it names the abstract operation from the `localCloudManagement` SD document, as well as input and output types. The examples use the console connector; on the Arrowhead connector the same request carries the identity token instead of the cookie.

For `POST`, `PUT`, `PATCH` and `DELETE` the header `X-Arrowhead-Action: confirm` is required (428 `"Missing action confirmation header"`). Request bodies are `application/json` and at most 64 KiB (larger: 500 `"Request body is too large"`; malformed: 500 `"Invalid JSON body"`); an empty body counts as `{}`. Responses are `application/json` with `Cache-Control: no-store`. Errors of the routes are answered with `{"error": "<text>"}` and the status codes `400 Bad Request` (invalid input), `401 Unauthorized`, `403 Forbidden`, `404 Not Found` (`"Not found"` for an unknown route), `405 Method Not Allowed`, `409 Conflict` (stored entry changed since it was read), `415 Unsupported Media Type` (login without JSON), `428 Precondition Required`, `429 Too Many Requests` (login locked), `500 Internal Server Error`, `502 Bad Gateway` (a provider request failed) and `503 Service Unavailable` (no authorized provider found by orchestration, or no console password). An error of a provider is passed on with the provider's status and error message.

2.1 GET /api/session

Operation: session
Output: Session

Whether the request carries a valid console session (no session needed). Console connector only.

A successful call returns `200 OK` with the output in the body.

```
GET /api/session HTTP/1.1
Host: 127.0.0.1:8490
Cookie: ah_console=<session>
```

Listing 2: An session invocation.

```
{
  "authenticated": true,
  "configured": true
}
```

Listing 3: An session response.

2.2 POST /api/session

Operation: login
Input: Login

Logs in with the console password. The body must be `application/json` (415 otherwise). On success the response sets `ah_console=<token>; HttpOnly; SameSite=Strict; Path=/; Max-Age=28800`. A wrong password is 401 `{"error": "Wrong password"}`; a locked client is 429 with `Retry-After` and `{"error": "Too many failed logins; try again later"}`. Other methods on this path are 405. Console connector only.

A successful call returns `204 No Content`.

```
POST /api/session HTTP/1.1
Host: 127.0.0.1:8490
Content-Type: application/json

```

```
{
  "password": "<console password>"
}
```

Listing 4: An login invocation.

2.3 DELETE /api/session

Operation: logout
Output: Session

Ends the console session; the response clears the cookie (`Max-Age=0`). Console connector only.

A successful call returns `200 OK` with the output in the body.

```
DELETE /api/session HTTP/1.1
Host: 127.0.0.1:8490
Cookie: ah_console=<session>
```

Listing 5: An logout invocation.

```
{
  "authenticated": false
}
```

Listing 6: An logout response.

2.4 GET /health

Operation: health
Output: Health

Liveness and ServiceRegistry registration state; no session needed. Console connector only (404 on the Arrowhead connector).

A successful call returns `200 OK` with the output in the body.

```
GET /health HTTP/1.1
Host: 127.0.0.1:8490
```

Listing 7: An health invocation.

```
{
  "status": "ok",
  "system": "LocalCloudManager",
  "registration": {
    "registered": true,
    "instanceId": "LocalCloudManager|generalManagement|1.0.0",
    "instanceIds": ["LocalCloudManager|generalManagement|1.0.0",
      "LocalCloudManager|localCloudManagement|1.0.0"],
    "error": null,
    "checkedAt": "2026-11-09T10:00:00.000Z"
  }
}
```

Listing 8: An health response.

2.5 GET /api/features

Operation: features
Output: Features

Which optional console views are enabled (not registered).

A successful call returns `200 OK` with the output in the body.

```
GET /api/features HTTP/1.1
Host: 127.0.0.1:8490
Cookie: ah_console=<session>
```

Listing 9: An features invocation.

```
{
  "localCloudConfiguration": true,
  "chipletFloorplans": false,
  "managerName": "LocalCloudManager"
}
```

Listing 10: An features response.

2.6 GET /api/overview

Operation: overview
Output: View

Overview: containers of the managed Compose project (Docker proxy), ServiceRegistry inventory with runtime state, registry management, naming check, CRL validity, brokers and sysMonitor flows, the last 60 samples and the registration state. A failing source is reported in `errors` and the corresponding `cloud` flag.

A successful call returns `200 OK` with the output in the body.

```
GET /api/overview HTTP/1.1
Host: 127.0.0.1:8490
Cookie: ah_console=<session>
```

Listing 11: An overview invocation.

```
{
  "collectedAt": "2026-11-09T10:00:00.000Z",
  "cloud": {"name": "arrowhead-v52-test-cloud", "release": "5.2.2",
            "registryReachable": true, "dockerReachable": true},
  "summary": {"at": "...", "running": 21, "containers": 23,
              "registeredSystems": 19, "registeredServices": 64,
              "unhealthy": 0, "incidents": 0, "unregisteredServices": []},
  "devices": [ ... ], "systems": [ ... ], "services": [ ... ],
  "registryManagement": { ... },
  "naming": {"valid": true, "violations": []},
  "revocation": {"thisUpdate": "...", "nextUpdate": "...",
                 "hoursRemaining": 70, "state": "ok"},
  "infrastructure": [ ... ], "orchestrations": [ ... ],
  "containers": [{"id": "0123456789ab", "name": "...", "service": "sysmonitor",
                  "displayName": "SysMonitor", "image": "...",
                  "state": "running", "status": "Up 2 hours (healthy)",
                  "health": "healthy", "created": "...", "managed": true}],
  "history": [ ... ],
  "registration": { ... },
  "errors": {"docker": null, "arrowhead": null, "naming": null}
}
```

Listing 12: An overview response.

2.7 GET /api/service-registry-management

Operation: serviceRegistryManagement
Output: View

Registry inventory grouped by device: systems with their services and runtime state, runtime-only containers and a summary.

A successful call returns `200 OK` with the output in the body.

```
GET /api/service-registry-management HTTP/1.1
Host: 127.0.0.1:8490
Cookie: ah_console=<session>
```

Listing 13: An serviceRegistryManagement invocation.

```
{
  "collectedAt": "...",
  "cloud": { ... },
  "devices": [{"name": "Device1", "addresses": ["10.0.0.1"], "metadata": {},
               "registered": true, "systems": [ ... ],
               "deployed": true, "serviceCount": 5}],
  "runtimeOnly": [ ... ],
  "summary": {"registeredDevices": 1, "deployedDevices": 1,
              "registeredSystems": 19, "deployedSystems": 18,
              "registeredServices": 64, "deployedServices": 60,
              "runtimeOnly": 0},
  "errors": { ... }
}
```

Listing 14: An serviceRegistryManagement response.

2.8 GET /api/datamanager?limit=*{limit}*

Operation: dataManager
Output: View

DataManager `dataHistorian` records of all systems and services (latest `limit` entries, 1 to 1000, default 240), the latest entry per measurement and numeric series.

Parameters: *limit*.

A successful call returns `200 OK` with the output in the body.

```
GET /api/datamanager?limit=240 HTTP/1.1
Host: 127.0.0.1:8490
Cookie: ah_console=<session>
```

Listing 15: An dataManager invocation.

```
{
  "collectedAt": "...",
  "count": 1,
  "latest": [{"id": "RandomDataProducer|randomNumberProducer|urn:...:value|1000",
              "name": "urn:arrowhead:rdp:value", "time": 1000, "unit": null,
              "value": {"v": 42}, "record": { ... },
              "system": "RandomDataProducer", "service": "randomNumberProducer"}],
  "entries": [ ... ],
  "series": [{"name": "urn:arrowhead:rdp:value",
              "points": [{"time": 1000, "value": 42, "unit": null}],
              "system": "RandomDataProducer", "service": "randomNumberProducer",
              "sourceCloud": null,
              "label": "RandomDataProducer . value - local"}]
```

```
}
```

Listing 16: An dataManager response.

2.9 GET /api/consumers

Operation: consumers
Output: View

Per provider service, the authorized consumers (ConsumerAuthorization policies) and the orchestrated consumers (DSO history, GateTunnel sessions, SimpleStore rules, DSO plant rules).

A successful call returns `200 OK` with the output in the body.

```
GET /api/consumers HTTP/1.1
Host: 127.0.0.1:8490
Cookie: ah_console=<session>
```

Listing 17: An consumers invocation.

```
{
  "collectedAt": "...",
  "localCloud": "TestCloud",
  "errors": [],
  "rows": [{"provider": "ServiceRegistry", "serviceDefinition": "temperature",
            "providerCloud": "TestCloud",
            "authorized": [{"consumerCloud": "TestCloud", "level": "MGMT",
                            "consumers": "Consumer", "scoped": []}],
            "orchestrated": [{"consumer": "Consumer", "consumerCloud": "TestCloud",
                              "via": "SimpleStore rule (priority 1)"}]}]
}
```

Listing 18: An consumers response.

2.10 GET /api/intercloud

Operation: interCloud
Output: View

Neighbour clouds, relay links of Gatekeeper and GateTunnel, GateTunnel sessions and the inter-cloud publishing of RandomDataProducer.

A successful call returns `200 OK` with the output in the body.

```
GET /api/intercloud HTTP/1.1
Host: 127.0.0.1:8490
Cookie: ah_console=<session>
```

Listing 19: An interCloud invocation.

```
{
  "collectedAt": "...",
  "localCloud": "TestCloud",
  "neighbourClouds": ["GoCloud|Company"],
  "neighbourError": null,
  "relay": {"gatekeeper": {"connected": true}, "gatetunnel": {"connected": true}},
  "sessions": {"sessions": []},
  "producer": {"status": "publishing"}
}
```

Listing 20: An interCloud response.

2.11 GET /api/sysmonitor?system={system}&limit={limit}

Operation: sysMonitorHistory
Output: View

Systems known to SysMonitor (`sysMonitorHistory`) with their latest observations, and the observations of one system (`system`, default the first; `limit` 1 to 1000, default 120).

Parameters: *system*, *limit*.

A successful call returns `200 OK` with the output in the body.

```
GET /api/sysmonitor?system=DataManager&limit=120 HTTP/1.1
Host: 127.0.0.1:8490
Cookie: ah_console=<session>
```

Listing 21: An sysMonitorHistory invocation.

```
{
  "collectedAt": "...",
  "retentionDays": 7,
  "storedObservations": 3,
  "systems": [{"systemName": "DataManager", "observationCount": 3,
               "latest": {"echo": {"status": "alive"}}}],
  "selectedSystem": "DataManager",
  "observations": [ ... ]
}
```

Listing 22: An sysMonitorHistory response.

2.12 GET /api/sysmonitor/subscriptions

Operation: sysMonitorSubscriptions
Output: View

SysMonitor providers, subscriptions and AMQP intake state (`sysMonitorManagement`).

A successful call returns `200 OK` with the output in the body.

```
GET /api/sysmonitor/subscriptions HTTP/1.1
Host: 127.0.0.1:8490
Cookie: ah_console=<session>
```

Listing 23: An sysMonitorSubscriptions invocation.

```
{
  "collectedAt": "...",
  "defaultSubscription": ["echo"],
  "providers": [{"systemName": "DataManager"}],
  "subscriptions": [],
  "intake": {"connected": true}
}
```

Listing 24: An sysMonitorSubscriptions response.

2.13 PUT /api/sysmonitor/subscriptions/*{system}*

Operation: subscribeSysMonitor
Input: SysMonitorSubscriptionRequest
Output: View

Sets the SysMonitor subscription of a provider system (`[A-Za-z0-9_-]+`); `enabled` defaults to true. The response is SysMonitor's.

Parameters: *system*.

A successful call returns `200 OK` with the output in the body.

```
PUT /api/sysmonitor/subscriptions/DataManager HTTP/1.1
Host: 127.0.0.1:8490
Cookie: ah_console=<session>
X-Arrowhead-Action: confirm
Content-Type: application/json

{
  "observationTypes": ["echo", "status"],
  "enabled": true
}
```

Listing 25: An subscribeSysMonitor invocation.

```
{
  "systemName": "DataManager",
  "enabled": true
}
```

Listing 26: An subscribeSysMonitor response.

2.14 DELETE /api/sysmonitor/subscriptions/*{system}*

Operation: unsubscribeSysMonitor
Output: Removal

Removes the SysMonitor subscription of a provider system.

Parameters: *system*.

A successful call returns `200 OK` with the output in the body.

```
DELETE /api/sysmonitor/subscriptions/DataManager HTTP/1.1
Host: 127.0.0.1:8490
Cookie: ah_console=<session>
X-Arrowhead-Action: confirm
```

Listing 27: An unsubscribeSysMonitor invocation.

```
{
  "removed": "DataManager"
}
```

Listing 28: An unsubscribeSysMonitor response.

2.15 GET /api/plantdescriptions

Operation: plantDescriptions
Output: View

Plant descriptions (first 200) with the registered systems and services offered as choices; `registryError` when the inventory failed.

A successful call returns `200 OK` with the output in the body.

```
GET /api/plantdescriptions HTTP/1.1
Host: 127.0.0.1:8490
Cookie: ah_console=<session>
```

Listing 29: An plantDescriptions invocation.

```
{
  "collectedAt": "...",
  "entries": [{"id": 1, "plantDescription": "test", "active": false, ... }],
  "count": 1,
  "systems": ["Consumer", "ServiceRegistry"],
  "services": [{"provider": "ServiceRegistry", "serviceDefinition": "temperature",
                "interfaces": ["generic_https"]}],
  "registryError": null
}
```

Listing 30: An plantDescriptions response.

2.16 POST /api/plantdescriptions

Operation: createPlantDescription
Input: PlantDescription
Output: PlantDescription

Creates a plant description in the PlantDescriptionEngine; the body is forwarded.

A successful call returns `201 Created` with the output in the body.

```
POST /api/plantdescriptions HTTP/1.1
Host: 127.0.0.1:8490
Cookie: ah_console=<session>
X-Arrowhead-Action: confirm
Content-Type: application/json

{
  "plantDescription": "Line 1",
  "active": false,
  "include": [],
  "systems": [],
  "connections": []
}
```

Listing 31: An createPlantDescription invocation.

```
{
  "id": 2,
  "plantDescription": "Line 1",
  ...
}
```

Listing 32: An createPlantDescription response.

2.17 PUT /api/plantdescriptions/*{id}*

Operation: replacePlantDescription
Input: PlantDescription
Output: PlantDescription

Replaces a plant description (numeric identifier; 404 without one).
Parameters: *id*.
A successful call returns `200 OK`.

```
PUT /api/plantdescriptions/2 HTTP/1.1
Host: 127.0.0.1:8490
Cookie: ah_console=<session>
X-Arrowhead-Action: confirm
Content-Type: application/json

{ ... }
```

Listing 33: An replacePlantDescription invocation.

2.18 PATCH /api/plantdescriptions/*{id}*

Operation: updatePlantDescription
Input: PlantDescription
Output: PlantDescription

Updates a plant description.
Parameters: *id*.
A successful call returns `200 OK`.

```
PATCH /api/plantdescriptions/2 HTTP/1.1
Host: 127.0.0.1:8490
Cookie: ah_console=<session>
X-Arrowhead-Action: confirm
Content-Type: application/json

{
  "active": true
}
```

Listing 34: An updatePlantDescription invocation.

2.19 DELETE /api/plantdescriptions/*{id}*

Operation: removePlantDescription
Output: Removal

Removes a plant description.
Parameters: *id*.
A successful call returns `200 OK` with the output in the body.

```
DELETE /api/plantdescriptions/2 HTTP/1.1
Host: 127.0.0.1:8490
Cookie: ah_console=<session>
X-Arrowhead-Action: confirm
```

Listing 35: An removePlantDescription invocation.

```
{
  "removed": 2
}
```

Listing 36: An removePlantDescription response.

2.20 GET /api/orchestration/policies

Operation: authorizationPolicies
Output: PolicyView

Local-cloud management policies of ConsumerAuthorization (editable) and provider-owned local policies (read-only); not registered.

A successful call returns `200 OK` with the output in the body.

```
GET /api/orchestration/policies HTTP/1.1
Host: 127.0.0.1:8490
Cookie: ah_console=<session>
```

Listing 37: An authorizationPolicies invocation.

```
{
  "entries": [ ... ],
  "readOnlyEntries": [ ... ],
  "collectedAt": "..."
}
```

Listing 38: An authorizationPolicies response.

2.21 POST /api/orchestration/policies

Operation: grantAuthorizationPolicy
Input: PolicyChange
Output: ManagementPolicy

Grants a local management policy (201), or replaces the stored policy of the same provider and target when `expected` matches it (200). 400 for an invalid policy, 409 for an unregistered provider or service, a policy that exists without `expected`, a changed policy or duplicate targets, 502 when a replacement could not complete.

A successful call returns `201 Created or 200 OK` with the output in the body.

```
POST /api/orchestration/policies HTTP/1.1
Host: 127.0.0.1:8490
Cookie: ah_console=<session>
X-Arrowhead-Action: confirm
Content-Type: application/json

{
  "policy": {"provider": "ServiceRegistry", "targetType": "SERVICE_DEF",
             "target": "temperature",
             "defaultPolicy": {"policyType": "WHITELIST",
                               "policyList": ["Consumer"]}},
  "expected": null
}
```

Listing 39: An grantAuthorizationPolicy invocation.

```
{
```

```
  "instanceId": "MGMT|LOCAL|ServiceRegistry|SERVICE_DEF|temperature",
  ...
}
```

Listing 40: An grantAuthorizationPolicy response.

2.22 DELETE /api/orchestration/policies/*{instanceId}*

Operation: revokeAuthorizationPolicy
Input: PolicyChange
Output: Removal

Revokes a local management policy (URL-encoded instance identifier) when `expected` matches the stored policy: 404 unknown, 428 without `expected`, 409 changed.

Parameters: *instanceId*.

A successful call returns `200 OK` with the output in the body.

```
DELETE /api/orchestration/policies/MGMT%7CLOCAL%7CServiceRegistry%7CSERVICE_DEF%7Ctemperature HTTP/
    1.1
Host: 127.0.0.1:8490
Cookie: ah_console=<session>
X-Arrowhead-Action: confirm
Content-Type: application/json

{
  "expected": { ... }
}
```

Listing 41: An revokeAuthorizationPolicy invocation.

```
{
  "removed": true,
  "id": "MGMT|LOCAL|ServiceRegistry|SERVICE_DEF|temperature"
}
```

Listing 42: An revokeAuthorizationPolicy response.

2.23 GET /api/orchestration/rules

Operation: orchestrationRules
Output: View

SimpleStore rules of the registered consumers, the orchestration catalog (producers with their interoperable consumers) and the aggregated DynamicServiceOrchestration history.

A successful call returns `200 OK` with the output in the body.

```
GET /api/orchestration/rules HTTP/1.1
Host: 127.0.0.1:8490
Cookie: ah_console=<session>
```

Listing 43: An orchestrationRules invocation.

```
{
  "catalog": {"producers": [{"instanceId": "ServiceRegistry|temperature|1.0.0",
                             "provider": "ServiceRegistry",
                             "serviceDefinition": "temperature",
                             "version": "1.0.0", "interfaces": [ ... ],
                             "compatibleConsumers": ["Consumer"]}],
```

```
                "consumers": [{"name": "Consumer", "version": null, "x509": true,
                               "interfaceTemplates": ["generic_https"]}]},
  "rules": [{"id": "123e4567-...", "consumer": "Consumer",
             "serviceInstanceId": "ServiceRegistry|temperature|1.0.0", "priority": 1}],
  "count": 1,
  "dynamic": [{"consumer": "Consumer", "serviceDefinition": "temperature",
               "count": 1, "kind": "local",
               "providers": [{"provider": "ServiceRegistry", "cloud": null}], ... }],
  "dynamicError": null,
  "collectedAt": "..."
}
```

Listing 44: An orchestrationRules response.

2.24 POST /api/orchestration/rules

Operation: createOrchestrationRule
Input: OrchestrationRule
Output: View

Creates a SimpleStore rule after checking that the producer is still offered and the consumer is interoperable with it (400 invalid, 409 stale or incompatible). The response is SimpleStore's.

A successful call returns `201 Created`.

```
POST /api/orchestration/rules HTTP/1.1
Host: 127.0.0.1:8490
Cookie: ah_console=<session>
X-Arrowhead-Action: confirm
Content-Type: application/json

{
  "consumer": "Consumer",
  "serviceInstanceId": "ServiceRegistry|temperature|1.0.0",
  "priority": 1
}
```

Listing 45: An createOrchestrationRule invocation.

2.25 PUT /api/orchestration/rules/*{id}*

Operation: changeOrchestrationRulePriority
Input: RulePriorityChange
Output: View

Changes the priority of a SimpleStore rule with `modify-priorities`, or by create and remove when `lcm.orchestration-priority-mode=replace` (not registered).

Parameters: *id*.

A successful call returns `200 OK`.

```
PUT /api/orchestration/rules/123e4567-e89b-12d3-a456-426614174000 HTTP/1.1
Host: 127.0.0.1:8490
Cookie: ah_console=<session>
X-Arrowhead-Action: confirm
Content-Type: application/json

{
  "priority": 2,
  "expected": {"consumer": "Consumer",
               "serviceInstanceId": "ServiceRegistry|temperature|1.0.0",
               "priority": 1}
}
```

Listing 46: An changeOrchestrationRulePriority invocation.

2.26 DELETE /api/orchestration/rules/*{id}*

Operation: removeOrchestrationRule
Output: Removal

Removes a SimpleStore rule (UUID; 400 for another identifier).
Parameters: *id*.
A successful call returns `200 OK` with the output in the body.

```
DELETE /api/orchestration/rules/123e4567-e89b-12d3-a456-426614174000 HTTP/1.1
Host: 127.0.0.1:8490
Cookie: ah_console=<session>
X-Arrowhead-Action: confirm
```

Listing 47: An removeOrchestrationRule invocation.

```
{
  "id": "123e4567-e89b-12d3-a456-426614174000",
  "removed": true,
  "at": "..."
}
```

Listing 48: An removeOrchestrationRule response.

2.27 GET /api/logs?service=*{service}*&tail=*{tail}*

Operation: logs
Output: LogLines

The last `tail` (1 to 500, default 150) log lines of a container of the managed project, through the Docker proxy (404 for an unknown service).
Parameters: *service*, *tail*.
A successful call returns `200 OK` with the output in the body.

```
GET /api/logs?service=sysmonitor&tail=150 HTTP/1.1
Host: 127.0.0.1:8490
Cookie: ah_console=<session>
```

Listing 49: An logs invocation.

```
{
  "service": "sysmonitor",
  "lines": ["2026-11-09T10:00:00.000000000Z ..."],
  "collectedAt": "..."
}
```

Listing 50: An logs response.

2.28 POST /api/containers/*{service}*/*{action}*

Operation: lifecycle
Output: LifecycleResult

Starts, stops or restarts (`start`, `stop`, `restart`) a container of a service in `lcm.managed-services` through the Docker proxy: 403 outside the allowlist, 404 unknown service.

Parameters: *service*, *action*.

A successful call returns `202 Accepted` with the output in the body.

```
POST /api/containers/sysmonitor/restart HTTP/1.1
Host: 127.0.0.1:8490
Cookie: ah_console=<session>
X-Arrowhead-Action: confirm
```

Listing 51: An lifecycle invocation.

```
{
  "service": "sysmonitor",
  "action": "restart",
  "accepted": true,
  "at": "..."
}
```

Listing 52: An lifecycle response.

2.29 GET /api/sos

Operation: sosConfigurations
Output: View

SoS compositions (all plant descriptions, up to 2000), shaped like plantDescriptions; `/api/sos/{id}` returns one composition (404 `"SoS not found"`); another path is 404 `"Unknown SoS operation"`. Not registered.

A successful call returns `200 OK`.

```
GET /api/sos HTTP/1.1
Host: 127.0.0.1:8490
Cookie: ah_console=<session>
```

Listing 53: An sosConfigurations invocation.

2.30 POST /api/sos

Operation: createSoS
Input: SoSChange
Output: PlantDescription

Stores a validated SoS composition (400 with the validation messages; 428 for an active composition without `activationConfirmed`).

A successful call returns `201 Created`.

```
POST /api/sos HTTP/1.1
Host: 127.0.0.1:8490
Cookie: ah_console=<session>
X-Arrowhead-Action: confirm
Content-Type: application/json

```

```
{
  "configuration": {"plantDescription": "Line 1", "active": false,
                    "include": [], "systems": [], "connections": []}
}
```

Listing 54: An createSoS invocation.

2.31 PUT /api/sos/*{id}*

Operation: replaceSoS
Input: SoSChange
Output: PlantDescription

Replaces an SoS composition when `expected` matches the stored one (409 otherwise; 400 when it includes itself).

Parameters: *id*.

A successful call returns `200 OK`.

```
PUT /api/sos/1 HTTP/1.1
Host: 127.0.0.1:8490
Cookie: ah_console=<session>
X-Arrowhead-Action: confirm
Content-Type: application/json

{
  "configuration": { ... },
  "expected": {"id": 1, "updatedAt": "...", ... }
}
```

Listing 55: An replaceSoS invocation.

2.32 DELETE /api/sos/*{id}*

Operation: removeSoS
Input: SoSChange
Output: Removal

Removes an SoS composition when `expected` matches the stored one.

Parameters: *id*.

A successful call returns `200 OK` with the output in the body.

```
DELETE /api/sos/1 HTTP/1.1
Host: 127.0.0.1:8490
Cookie: ah_console=<session>
X-Arrowhead-Action: confirm
Content-Type: application/json

{
  "expected": { ... }
}
```

Listing 56: An removeSoS invocation.

```
{
  "removed": 1
}
```

Listing 57: An removeSoS response.

2.33 GET /api/localcloud/*{route}*

Operation: localCloudConfiguration
Input: Object
Output: View

Local-cloud configuration (404 "`Local cloud configurator is not enabled`" when switched off). The route selects one operation of `localCloudConfiguration` of LocalCloudConfigurator [6], consumed through orchestration; the request body is forwarded and status and body are returned unchanged. Other routes are 404 "`Unknown local cloud configuration operation`".

Parameters: *route*.

A successful call returns `the configurator's status` with the output in the body.

```
POST /api/localcloud/plan HTTP/1.1
Host: 127.0.0.1:8490
Cookie: ah_console=<session>
X-Arrowhead-Action: confirm
Content-Type: application/json

{
  "systems": ["timemanager"],
  "remove": [],
  "reapply": false,
  "smoke": true
}
```

Listing 58: An localCloudConfiguration invocation.

```
{
  "plan": {"cloud": "TestCloud", "catalogHash": "...", "systems": [ ... ],
           "steps": [ ... ], "notes": [ ... ]},
  "hash": "..."
}
```

Listing 59: An localCloudConfiguration response.

2.34 GET /api/chiplet/floorplans

Operation: chipletFloorplans
Input: Object
Output: View

Chiplet floorplan view when `lcm.chiplet-floorplans.enabled=true` (404 otherwise), through orchestrated ChipletFloorplanGenerator services: `GET` list (`page`, `size` up to 200), `/library` and `/{id}`; `POST` create (stored inactive, 201) and `POST /check-network`; `PUT /{id}` with `expectedUpdatedAt` (409 when changed); `DELETE /{id}`. Bodies carry `floorplan` with 1–64 chiplets and at most 1024 links. Not registered.

A successful call returns `200 OK`.

```
GET /api/chiplet/floorplans?page=0&size=200 HTTP/1.1
Host: 127.0.0.1:8490
Cookie: ah_console=<session>
```

Listing 60: An chipletFloorplans invocation.

3 Data Models

Here, all data objects that can be part of the service calls associated with this service are listed in alphabetic order. Note that each subsection, which describes one type of object, begins with the *struct* keyword, which is meant to denote a JSON Object that must contain certain fields, or names, with values conforming to explicitly named types. Fields that are not needed may be omitted or `null` in requests; empty fields are omitted in responses. As a complement to the primary types defined in this section, there is also a list of secondary types in Section 3.17, which are used to represent things like hashes, identifiers and texts.

3.1 struct LifecycleResult

Object Field	Value Type	Description
"service"	String	Compose service.
"action"	String	`start`, `stop` or `restart`.
"accepted"	Boolean	Forwarded to the Docker proxy.
"at"	String	Time of the action.

3.2 struct LogLines

Object Field	Value Type	Description
"service"	String	Compose service.
"lines"	Array<String>	Log lines (`tail` latest).
"collectedAt"	DateTime	Collection time.

3.3 struct Health

Field	Type	Description
status	String	`ok`.
system	Name	`LocalCloudManager`.
registration	Object	`registered`, `instanceId`, `instanceIds`, `error`, `checkedAt`.

3.4 struct Login

Object Field	Value Type	Description
"password"	String	Console password (JSON only; 204 on success, 401 wrong, 429 locked).

3.5 struct OrchestrationRule

Object Field	Value Type	Description
"consumer"	String	Consumer system.
"serviceInstanceId"	String	Service instance.
"priority"	Number	Priority (1 highest).

3.6 struct PlantDescription

Object Field	Value Type	Description
"*"	Object	Plant description as in PlantDescriptionEngine `plantDescriptionManagement`.

3.7 struct Session

Object Field	Value Type	Description
"authenticated"	Boolean	Whether the request carries a valid console session.
"configured"	Boolean	Whether a console password is configured.

3.8 struct SysMonitorSubscriptionRequest

Object Field	Value Type	Description
"observationTypes"	Array<String>	Types to receive, `*` for all.
"enabled"	Boolean	Default true.

3.9 struct View

Object Field	Value Type	Description
"*"	Object	View document assembled from the providing systems; a failing source is reported in the document instead of failing the request.

3.10 struct Features

Object Field	Value Type	Description
"localCloudConfiguration"	Boolean	Local-cloud configuration view enabled.
"chipletFloorplans"	Boolean	Chiplet floorplan view enabled.
"managerName"	String	LocalCloudManager, or ChipletLocalCloudManager with floorplans.

3.11 struct ManagementPolicy

Object Field	Value Type	Description
"instanceId"	String	Policy instance (assigned by Consumer Authorization).
"provider"	String	Registered local provider.
"targetType"	String	`SERVICE_DEF` or `EVENT_TYPE`.
"target"	String	Service definition or event type.
"description"	String	At most 1000 characters.
"defaultPolicy"	Object	`policyType` (`ALL`, `WHITELIST`, `BLACKLIST`) and `policyList` of local system names.
"scopedPolicies"	Object	Per operation, a policy like `defaultPolicy` (at most 100).

3.12 struct PolicyChange

Object Field	Value Type	Description
"policy"	ManagementPolicy	Policy to grant (POST).
"expected"	ManagementPolicy	Stored policy as last read; required to replace or revoke.

3.13 struct PolicyView

Object Field	Value Type	Description
"entries"	Array	Editable local management policies.
"readOnlyEntries"	Array	Provider-owned local policies.
"collectedAt"	DateTime	Collection time.

3.14 struct Removal

Object Field	Value Type	Description
"removed"	Object	Identifier of, or `true` for, the removed entry.
"id"	String	Removed policy (policies only).

3.15 struct RulePriorityChange

Object Field	Value Type	Description
"priority"	Number	New priority, 1 to 2147483647.
"expected"	OrchestrationRule	Rule as last read (409 if changed, 428 if missing).

3.16 struct SoSChange

Object Field	Value Type	Description
"configuration"	Object	SoS model: `plantDescription` (name), `active`, `include`, `systems` with ports, `connections`.
"expected"	Object	Stored composition as last read (replace and remove).
"activationConfirmed"	Boolean	Must be `true` to save an active composition (428 otherwise).

3.17 Primitives

As all messages are encoded using the JSON format [4], the following primitive constructs, part of that standard, become available. Note that the official standard is defined in terms of parsing rules, while this list only concerns syntactic information. Furthermore, the Object and Array types are given optional generic type parameters, which are used in this document to signify when pair values or elements are expected to conform to certain types.

Type	Description
Array	A JSON Array [4]; Array<A> has elements of type A.
Boolean	JSON Boolean, `true` or `false`.
DateTime	JSON String with an ISO 8601 UTC timestamp, e.g. `2026-10-22T12:00:00Z`.
Name	JSON String, an Arrowhead system name.
Number	JSON Number, an integer.
Object	JSON Object; Object<A> has string keys and values of type A.
String	JSON String.

4 References

[1] R. Fielding, M. Nottingham, and J. Reschke, “Http/1.1,” IETF, RFC 9112, 2022.

[2] E. Rescorla, “The transport layer security (tls) protocol version 1.3,” IETF, RFC 8446, 2018.

[3] T. Dierks and E. Rescorla, “The transport layer security (tls) protocol version 1.2,” IETF, RFC 5246, 2008.

[4] T. Bray, “The javascript object notation (json) data interchange format,” IETF, RFC 8259, 2017.

[5] J. Delsing, “localCloudManagement – Service Description (LocalCloudManager),” Eclipse Arrowhead, SD 5.2.2, 2026, localCloudManager-v5.2.2-localCloudManagement-SD.pdf.

[6] ——, “localCloudConfiguration HTTP/TLS/JSON – Interface Design Description (LocalCloudConfigurator),” Eclipse Arrowhead, IDD 5.2.2, 2026, localCloudConfigurator-v5.2.2-localCloudConfiguration-HTTP-IDD.pdf.

5 Revision History

5.1 Amendments

No.	Date	Version	Subject of Amendments	Author
1	2026-10-22	5.2.2	Document created from the Arrowhead LaTeX templates for the 5.2.2 Java Spring implementation, including the local extensions of this workspace.	Jerker Delsing
2	2026-10-30	5.2.2	Checked against the primary Java source tree (Eclipse-Arrowhead-java): console operations outside the registered list (features, authorization policies, rule priority, SoS, local-cloud configuration, chiplet floorplans) added; success responses of logout, unsubscribeSysMonitor, removePlantDescription and removeOrchestrationRule corrected; error codes completed.	Claude Opus 5.5 (draft)
3	2026-11-09	5.2.2	Java-Spring implementation: Arrowhead connector 8491 (`generic_https`, `CERT_AUTH`, identity token, grant check) and console connector 8490 (session cookie, Host/Origin, proxy secret); every route with method, path, status codes and JSON from the controller; session operations console-only; `/health`; local-cloud configuration through the orchestrated `localCloudConfiguration` service. Node.js/nginx descriptions removed.	Claude Opus 5.5 (draft)

5.2 Quality Assurance

No.	Date	Version	Approved by
1		5.2.2	

sysMonitor AMQP/TLS/JSON

Interface Design Description

Service ID: *"sysMonitor"*

Abstract
This document describes the AMQP 0-9-1 protocol with TLS payload security and JSON payload encoding variant of the `sysMonitor` service of the LocalCloudManager system (SysMonitor-AMQP-IDD-5.2.2).

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1 Overview

This document describes the AMQP binding of the native `sysMonitor` service of LocalCloudManager.
It's implemented using protocol, encoding as stated in below table:

Profile type	Type	Version
Transfer protocol	AMQP	0-9-1 [1]
Data encryption	TLS	1.3 [2], 1.2 [3]
Encoding	JSON	RFC 8259 [4]
Compression	N/A	-
Semantics	N/A	-
Ontology	N/A	-

Table 1: Communication and semantics details used for the `sysMonitor` service interface

This document provides the Interface Design Description IDD to the *sysMonitor – Service Description* document [5].

1.1 Interface registration

With `sysmonitor.enabled=true`, the common sysMonitor publisher of LocalCloudManager registers the service at start-up with interface template `generic_amqps` (protocol `amqps`, policy `CERT_AUTH`; `generic_amqp` without TLS) and the following properties and metadata:

```
{
  "serviceDefinitionName": "sysMonitor",
  "version": "1.0.0",
  "metadata": {
    "contract": "SysMonitor-AMQP-IDD-5.2.2",
    "schemaVersion": "1.0",
    "publisherMode": "native",
    "observationTypes": [
      "echo",
      "power",
      "status"
    ],
    "delivery": "persistent, publisher-confirmed"
  },
  "interfaces": [
    {
      "templateName": "generic_amqps",
      "protocol": "amqps",
      "policy": "CERT_AUTH",
      "properties": {
        "accessAddresses": [
          "amqp-broker"
        ],
        "accessPort": 5671,
        "exchange": "arrowhead",
        "baseRoutingKey": "TestCloud.sysmonitor.LocalCloudManager.LocalCloudManager",
        "operations": [
          "echo",
          "power",
          "status"
        ]
      }
    }
  ]
}
```

Listing 1: Registered service instance (TestCloud).

2 Service Operations

The publisher is the `SysMonitorPublisher` of the vendored Eclipse Arrowhead common library (RabbitMQ Java client), configured with the `sysmonitor.*` Spring properties (`enabled`, `broker.address`, `broker.port`, `local.cloud`, `service.instance`, `publish.interval`, `ssl.*`). The echo health comes from the Docker health indicator: degraded while the Docker proxy cannot be reached; `serviceCount` is the number of services in the system information (2). The publisher connects to the broker (default `amqp-broker`, port 5671) with TLS, verifies the broker host name and authenticates with its X.509 system certificate (SASL `EXTERNAL`; the certificate CN is the broker user name). Every publication interval it publishes each observation to the topic exchange `arrowhead` with the routing key `<baseRoutingKey>.<operation>`, where the base routing key is `<Cloud>.sysmonitor.<System>.<serviceInstance>`. Messages are persistent (delivery mode 2), `application/json`, carry the observation identifier as message id, and are publisher-confirmed (timeout 5 s); a failed cycle is retried with a new connection at the next interval. The broker's topic permissions allow the system to publish only below its own base routing key.

2.1 PUBLISH `<Cloud>.sysmonitor.LocalCloudManager.LocalCloudManager.echo`

Operation: echo
Output: Observation

Liveness: `status` (`alive` or `degraded`), `uptimeSeconds` and a `detail` text; quality `good` or `uncertain`.

2.2 PUBLISH `<Cloud>.sysmonitor.LocalCloudManager.LocalCloudManager.power`

Operation: power
Output: Observation

Power supply: `supplyType=line`, `state=available`; unit `1`.

2.3 PUBLISH `<Cloud>.sysmonitor.LocalCloudManager.LocalCloudManager.status`

Operation: status
Output: Observation

Process state: `state=running`, `health` (`healthy`/`degraded`) and `serviceCount`, the number of registered services.

```
{
  "schemaVersion": "1.0",
  "observationId": "LocalCloudManager-echo-1792670400000-42",
  "providerSystem": "LocalCloudManager",
  "serviceInstance": "LocalCloudManager",
  "observationType": "echo",
  "sourceTime": "2026-10-22T12:00:00.000Z",
  "sequence": 42,
  "value": {
    "status": "alive",
    "uptimeSeconds": 3600,
    "detail": "LocalCloudManager is running"
  },
  "quality": "good",
  "metadata": {
```

```
16      "publisherMode": "native"
17    }
18  }
```

Listing 2: An echo observation.

3 Data Models

3.1 struct Observation

Object Field	Value Type	Description
"schemaVersion"	String	`1.0`.
"observationId"	String	`<System>-<type>-<epoch ms>-<sequence>`.
"providerSystem"	String	The publishing system.
"serviceInstance"	String	Service instance, default the system name.
"observationType"	String	`echo`, `power`, `status`.
"sourceTime"	String	ISO 8601 UTC time with milliseconds.
"sequence"	Number	Sequence number per publisher.
"value"	Object	Observed value, see the operations.
"unit"	String	`1` for `power` and `status`; absent for `echo`.
"quality"	String	`good` or `uncertain`.
"metadata"	Object	`{"publisherMode": "native"}`.

3.2 Primitives

Type	Description
Number	JSON Number, an integer.
Object	JSON Object; Object<A> has string keys and values of type A.
String	JSON String.

4 References

[1] AMQP Working Group, “Amqp 0-9-1 specification,” AMQP Working Group, Tech. Rep., 2008.

[2] E. Rescorla, “The transport layer security (tls) protocol version 1.3,” IETF, RFC 8446, 2018.

[3] T. Dierks and E. Rescorla, “The transport layer security (tls) protocol version 1.2,” IETF, RFC 5246, 2008.

[4] T. Bray, “The javascript object notation (json) data interchange format,” IETF, RFC 8259, 2017.

[5] J. Delsing, “sysMonitor – Service Description (LocalCloudManager),” Eclipse Arrowhead, SD 5.2.2, 2026, localCloudManager-v5.2.2-sysMonitor-SD.pdf.

5 Revision History

5.1 Amendments

No.	Date	Version	Subject of Amendments	Author
1	2026-10-22	5.2.2	Document created from the Arrowhead LaTeX templates for the 5.2.2 Java Spring implementation, including the local extensions of this workspace.	Jerker Delsing
2	2026-10-30	5.2.2	Checked against the primary Java source tree (Eclipse-Arrowhead-java): services, operations, paths and data models unchanged.	Claude Opus 5.5 (draft)
3	2026-11-09	5.2.2	Java-Spring implementation: the observations are published by the common-library `SysMonitorPublisher` (RabbitMQ Java client) with the `sysmonitor.*` properties; health from the Docker proxy; wire format unchanged.	Claude Opus 5.5 (draft)

5.2 Quality Assurance

No.	Date	Version	Approved by
1		5.2.2	

1 HTTPS interface overview

Target secure interface `generic_https`, policy CERT_AUTH at transport; default standalone origin is `https://<provisioned-host>:8480`. The UI and API share the listener. Configure valid SANs and reverse-proxy client certificates if a proxy is used. Requests/responses are JSON except assets, echo and204login. Documented routes are the actual handlers; source registration metadata has known method/path mismatches. Tables use grouped methods or aliases where indicated.

Method	Actual handler path	Access / behavior
GET	`/health`	transport. Runtime configuration, not readiness proof
GET	`/and/ui`	transport. Static embedded index
GET	`/app.jsand/assets/app.js`	transport. Embedded JavaScript
GET	`/styles.cssand/assets/styles.css`	transport. Embedded CSS
GET	`/api/session`	console policy. authenticated/configured booleans
POST	`/api/session`	console policy. Password login; 204 and cookie
DELETE	`/api/session`	console policy. Invalidate cookie/session
GET	`/api/overview`	session. Inventory/runtime aggregation; synthetic flags noted
GET	`/api/service-registry-management`	session. Grouped devices/systems/services/runtime
GET	`/api/datamanager`	session. Flatten historian; limit default240, max1000
GET	`/api/consumers`	session. Service and simple-store consumer view
GET	`/api/intercloud`	session. Static unavailable placeholder
GET	`/api/sysmonitor`	session. Inherited observation adapter; optional system query
GET	`/api/sysmonitor/subscriptions`	session. Read subscription adapter; synthetic intake flags
PUT, DELETE	`/api/sysmonitor/subscriptions/{providerSystem}`	session + confirm. Acknowledgement only; NO provider write
GET	`/api/plantdescriptions`	session. Plant records and registry catalogue
POST	`/api/plantdescriptions`	session + confirm. Create through provider adapter
PUT, PATCH, DELETE	`/api/plantdescriptions/{id}`	session + confirm. Provider replace/update/remove
GET	`/api/orchestration/rules`	session. Simple store plus producer/consumer catalogue
POST	`/api/orchestration/rules`	session + confirm. Validate consumer/instance/priority then create

Method	Actual handler path	Access / behavior
DELETE	`/api/orchestration/rules/{id}`	session + confirm. Remove via simple-store adapter
GET	`/api/logs`	session. service query; tail default150, max500
POST	`/api/containers/{service}/{action}`	session + confirm. Static allowlist, project filter; start/stop/restart
GET	`/localcloud/management/status`	Arrowhead management. Legacy status response
GET	`/localcloud/management/systems`	Arrowhead management. Registry systems query
GET	`/localcloud/management/services`	Arrowhead management. Registry instances query
GET	`/localcloud/monitor/echo`	Arrowhead identity. Text Got it!; registry metadata differs

2 Session and mutation DTOs

`POST/api/session` body: `{"password":"..."}`. Success204with Set-Cookie `ah_console`, HttpOnly, SameSite=Strict, Path=/, Max-Age default28800; no Secure flag. `GET/api/session` responds authenticated/-configured; DELETE invalidates and expires the cookie. Missing hash503, wrong password401, lockout429with Retry-After; lockout follows five failures in five minutes and lasts one minute. Host or Origin denial403.

All `/api/` requests require allowed Host/Origin; all except session routes require a session. Mutation confirmation is exactly `X-Arrowhead-Action:confirm`; absent confirmation428. This is not cryptographic replay protection or origin-port isolation.

Rule creation body requires consumer, serviceInstanceId and positive integer priority. The adapter verifies producer availability and compatible consumer from registry data before forwarding. Successful create201; remove200with id/removed/at. Provider failures generally502with an error string; malformed rule JSON400.

Plant mutations forward JSON to the inherited provider path; create201, updates200, delete200with removed id. Subscription writes respond providerSystem/updated/removed but do not forward any change. Lifecycle responds202with service/action/accepted/at, and only start/stop/restart on a static allowlisted project container is accepted.

3 Read response DTOs

Overview includes collectedAt, cloud, summary, devices, systems, services, registryManagement, naming, infrastructure, orchestrations, containers, history, registration and errors. Grouping includes devices/runtimeOnly and registered/deployed counts. Some flags and empty collections are synthetic, not verified state.

Data view uses limit (default240,max1000), collectedAt/count/latest/entries/series; entries are flattened inherited historian data. Consumers returns localCloud/errors/rows with provider/definition/authorization/orchestration views; cloud labels are partly hard-coded TestCloud.

SysMonitor view includes collectedAt, retentionDays, storedObservations, systems, selectedSystem and observations; the optional query is system. Subscription read returns defaultSubscription/providers/subscriptions/intake, with synthetic connected/providerFound values. Plant read returns entries/systems/services/registryError.

Logs use service and tail query (default150,max500), returning service/lines/collectedAt. Runtime rows include id/name/service/displayName/image/state/status/health/created/managed. Logs may include any container returned by the Compose project filter, even when lifecycle management is disallowed.

4 Compatibility and release gates

This relocation intentionally preserves the latest located manager's business behavior. New changes are packaging (standalone nested module/role-specific launcher), embedded UI fallback, documentation and tests. This is not a security remediation or full Java-equivalence implementation.

1. `PUT/DELETE/api/sysmonitor/subscriptions/{providerSystem}` currently checks confirmation and returns updated/removed flags without forwarding a write. Do not interpret this as applied configuration. This remains a release blocker.

2. `/api/intercloud` returns a static unavailable/no-relay summary. It does not inspect Gatekeeper/GateTunnel. Consumer cloud labels include hard-coded TestCloud strings; no live foreign authorization catalogue is queried.

3. Overview hard-codes registration/naming flags and leaves several collections empty. Subscription status hard-codes intake.connected=true/providerFound=false. These are display adapters, not authoritative readiness or security evidence.

4. Historian, SysMonitor, plant and orchestration adapters target the inherited Go-cloud routes/shapes. They are not proven compatible with the newer generic Go support port or latest Java APIs. The standalone module must be qualified against the providers actually deployed.

5. Console Host/Origin checks compare host names, not exact origins/ports. The `ah_console` cookie is not port-isolated and lacks Secure. Sessions/history are in-memory and reset on restart. Missing password disables APIs. Login lockout is global, five failures in five minutes then one-minute lockout; this is not comprehensive abuse protection.

6. The manager API is dispatched before the shared application's token/management checks. Its boundary is console password/session plus the transport's mTLS if enabled, not a bearer-token-only API. Static assets and health have no console session requirement. A proxy must preserve this intended boundary.

7. Authentication has explicit CN/profile pinning. Other provider clients use CA/SAN validation but do not generally pin expected provider CN. Do not claim universal authority-substitution resistance for this inherited code. Redirects are disabled. Optional CRLs reload for new connections only; omitted CRLs mean no revocation enforcement.

8. Lifecycle uses the static systemToCompose service allowlist and Compose-project filtered container list; MANAGED_SERVICES is not consumed by this implementation. Action is start/stop/restart only. Logs, however, can read any returned project container, not just the lifecycle allowlist. The raw Docker socket still confers broad daemon privilege: application checks are not a least-privilege socket boundary. Production needs a constrained proxy and deployment review.

9. Shared runtime source includes other roles for compile-time dependencies and inherited tests. The dedicated executable refuses selecting those roles; they are not a second support-system deployment. No Docker socket or live state is needed by tests.

10. Telemetry AMQP publisher has no durable local outbox; retries/reconnects do not establish exactly-once delivery. power=line/available and process health are generated status, not measurements of physical hardware. Registration is retried every20seconds; displayed registration flags are not confirmation. No real broker test is claimed here.

11. Shared state is the inherited JSON format, not the newer support port's transactional state model. There is no new Java SQL migration, encryption-at-rest or multi-process ownership guarantee. Keep independent state and backups.

12. Historical Markdown/SysML claims of three-way interchangeability are retained only under docs/legacy. The revised model and new SysD/SysDD/SD/IDD set distinguish intended common contracts from actual Go qualification.

13. The shared serviceOperations helper generates POST/name metadata for localCloudManagement and the legacy monitor service, whereas actual handlers include GET/PUT/PATCH/DELETE and nested paths. Registration metadata is therefore not an authoritative HTTP contract. The new IDDs document actual handler methods/paths; correcting advertised metadata is separate functional work, not silently folded into relocation.

Source relocation does not modify running containers or remove their deployment source. Before switching deployment build paths, close these gaps, qualify provider DTOs/policies, review Docker access, isolate browser trust domains, and test rollback.

5 References and verification

Authoritative moved code: LocalCloudManager/internal/cloud and embedded ui sources. Exact original/relocated SHA256 hashes are in docs/source-manifest.json. docs/legacy preserves the earlier local Markdown/SysML,

not an endorsement of its parity claims. The revised SysML variability model is descriptive and has not been checked by a SysML parser. Executed checks and qualification limits are recorded in VERIFICATION.md. Source movement does not deploy or alter running clouds.

1 Legacy HTTPS echo

Request `GET/localcloud/monitor/echo`, no body. Secure mode requires a verified local certificate and valid bound Arrowhead identity token. Response200text `Gotit!`. Authentication failures401. No operator-console cookie is substituted for this legacy Arrowhead identity check. The shared handler accepts echo suffixes broadly; this is a compatibility route, not a strict operation inventory or canonical sysMonitor interface.

2 References and verification

Authoritative moved code: LocalCloudManager/internal/cloud and embedded ui sources. Exact original/relocated SHA256 hashes are in docs/source-manifest.json. docs/legacy preserves the earlier local Markdown/SysML, not an endorsement of its parity claims. The revised SysML variability model is descriptive and has not been checked by a SysML parser. Executed checks and qualification limits are recorded in VERIFICATION.md. Source movement does not deploy or alter running clouds.

1 AMQPS binding

AMQP0-9-1 overTLS, interface `generic_amqps`, policy CERT_AUTH. Broker AMQP_HOST/AMQP_PORT defaultamqp-broker:5671; SASL EXTERNAL with LocalCloudManager service certificate. Exchange `arrowhead`, routing `<Cloud>.sysmonitor.LocalCloudManager.LocalCloudManager.<observationType>`. Operations/types `echo`, `power`, `status`; content type application/json, delivery mode persistent, publisher confirm with a10second wait bound. Configure broker SANs, trust, exchange and ACLs independently.

2 Observation schema

Envelope fields: schemaVersion1.0, observationId, providerSystem, serviceInstance, observationType, sourceTime (UTC-Z), sequence, value, quality, metadata. metadata contains publisherMode=native and implementation=go. Non-echo observations include unit1.

Echo value: status=alive, uptimeSeconds, detail. Power: supplyType=line, state=available. Status: state=running, health=healthy, serviceCount. These values do not establish provider readiness or physical supply state.

Registry metadata contract SysMonitor-AMQP-IDD-5.2.2, schemaVersion1.0, publisherMode native, observationTypes array and delivery persistent/publisher-confirmed. baseRoutingKey is `cloud.sysmonitor.LocalCloudManager.LocalCloudManager`. No plaintext AMQP fallback is provided by this publisher.

3 References and verification

Authoritative moved code: LocalCloudManager/internal/cloud and embedded ui sources. Exact original/relocated SHA256 hashes are in docs/source-manifest.json. docs/legacy preserves the earlier local Markdown/SysML, not an endorsement of its parity claims. The revised SysML variability model is descriptive and has not been checked by a SysML parser. Executed checks and qualification limits are recorded in VERIFICATION.md. Source movement does not deploy or alter running clouds.

1 localCloudManagement HTTP interface

The actual listener is plain HTTP; the dedicated launcher defaults to `http://127.0.0.1:8480/`. The inherited registry descriptor uses generic_http/policy NONE. Do not label this as a native HTTPS/mTLS interface. An external proxy and its certificates are outside this source relocation, and cannot repair legacy provider authentication by themselves.

1.1 Actual handler inventory

Method	Path	Behavior
GET	`/health`	Text OK; not dependency readiness
GET	`/` and `/ui`	Embedded dashboard; no console session
GET	`/app.js`, `/assets/app.js`	Embedded JavaScript
GET	`/styles.css`, `/assets/styles.css`	Embedded CSS
GET	`/api/session`	authenticated/configured state
POST	`/api/session`	Password login, success204with cookie
DELETE	`/api/session`	Logout,200with expired cookie
GET	`/api/overview`	Registry/runtime aggregation
GET	`/api/service-registry-management`	Devices/systems/services/runtime grouping
GET	`/api/datamanager`	Historian adapter; limit1..1000, default240
GET	`/api/consumers`	Inherited partial consumer view
GET	`/api/intercloud`	Static unavailable placeholder
GET	`/api/sysmonitor`	Inherited observation adapter, optional system query
GET	`/api/sysmonitor/subscriptions`	Read adapter, synthetic intake flags
PUT, DELETE	`/api/sysmonitor/subscriptions/{provider}`	Confirmed acknowledgement ONLY; no write
GET	`/api/plantdescriptions`	Plant/registry catalogue
POST	`/api/plantdescriptions`	Confirmed provider create; currently200
PUT, PATCH	`/api/plantdescriptions/{id}`	Confirmed provider update; both forward PUT
DELETE	`/api/plantdescriptions/{id}`	Confirmed provider removal
GET	`/api/orchestration/rules`	Simple-store catalogue/rules
POST	`/api/orchestration/rules`	Confirmed validation then provider create
DELETE	`/api/orchestration/rules/{id}`	Confirmed provider removal
GET	`/api/logs`	Project-container logs, service/tail query
POST	`/api/containers/{service}/{action}`	Confirmed allowlisted project lifecycle

Method	Path	Behavior
GET	`/localcloud/management/status`	Separate legacy requester status
GET	`/localcloud/management/systems`	Separate legacy management registry query
GET	`/localcloud/management/services`	Separate legacy management service query
GET	`/localcloud/monitor/echo`	Separate legacy echo

1.2 Session and access

All `/api/` requests require allowed Host/Origin host names and a configured password hash. Non-session routes also require a valid console cookie. Host-name matching ignores exact ports. Missing/invalid hash503, rejected Host/Origin403, absent/expired session401. Static UI/health and legacy routes are not covered by this console middleware.

POST session accepts JSON with password string. The configured hash is scrypt with N16384,r8,p1. A correct password returns204and sets `ah_console`, HttpOnly, SameSite=Strict, Path=/, Max-Age default28800. There is no Secure flag or port isolation. GET session returns authenticated/configured. DELETE removes the token and expires the cookie. Five wrong passwords in five minutes trigger one-minute lockout;429includes Retry-After60. Wrong password401.

Mutations require exactly `X-Arrowhead-Action:confirm`. Missing confirmation428with JSON error. Axum body/content-type extraction can reject malformed JSON before the handler checks confirmation. This header is not cryptographic replay protection.

1.3 Mutation DTOs

Rule creation requires consumer, serviceInstanceId and positive unsigned priority. Unknown/incompatible producer choice409; invalid required values400. The provider body wraps the candidate in a candidates array. Remove uses ids query at the inherited simple-store endpoint. Provider success is returned as JSON, generally200; provider/network failures502with error. No distributed transaction or atomic rollback is guaranteed.

Plant create/update JSON is forwarded to the inherited provider path. PATCH also forwards PUT. Create currently returns200. Delete forwards DELETE. Subscription PUT and DELETE ignore the body and return providerSystem/updated=true without applying any provider change; a200response is not configuration confirmation.

Lifecycle action must be start, stop or restart. A static source allowlist and Compose project filter constrain selection. Unknown action400, unlisted target403, unknown returned project service404, Docker failure502. Success JSON contains service/action/accepted=true/at and currently returns200, not202. The raw Docker socket privilege is not reduced by these application checks.

1.4 Read DTOs and errors

Overview includes collectedAt, cloud, summary, devices, systems, services, registryManagement, naming, infrastructure, orchestrations, containers, history, registration and errors. Some flags and collections are placeholders. Registry grouping uses devices, runtimeOnly and summary; no assurance of authoritative readiness follows from these fields.

Historian returns collectedAt/count/latest/entries/series; source data comes from inherited per-system/per-service SenML paths. Consumers uses TestCloud labels and empty authorized/orchestrated lists. Intercloud is static unavailable. SysMonitor reads group observations with sourceTime/value/quality, but upstream errors are hidden as empty data; subscription intake status is synthetic.

Logs accept service and tail (clamped1..500, default150). Response includes service/lines/collectedAt. Logs may include any container returned for the project, even if it is not lifecycle-allowed. Runtime rows include id/name/service/displayName/image/state/status/health/created/managed.

Console handler errors generally use a JSON error string. Legacy shared-runtime errors use errorMessage/errorCode/exceptionType/origin. Unknown routes404; unsupported methods405 through Axum. Exact Java DTO/error parity and a full browser deployment test are not established.

2 References and qualification

See `docs/source-manifest.json` for original/relocated source SHA256 records, `VERIFICATION.md` for checks actually run, and `docs/LocalCloudManager.sysml` for the descriptive, unparsed variability model. No Java interchangeability, native TLS or sysMonitor publisher is claimed. Existing cloud source and running deployment remain unchanged.

1 Legacy monitor HTTP interface

Request `GET/localcloud/monitor/echo`, no body. Response200text `Gotit!`. The source's legacy requester accepts `Authorization:BearerSYSTEM//name` as a declared name; alternative bearer handling is inherited and does not establish certificate-bound authentication. No native TLS client certificate is required by this listener. Invalid or absent legacy requester yields401.

This route is separate from `/api/` console-cookie middleware. Console cookies do not protect or replace the legacy requester check. An echo response is not evidence of registry publication, trusted identities, provider health, AMQP publishing or production suitability.

The source registers monitor1.0.0 at `/localcloud/monitor` with GET/echo, generic_http/policy NONE. No sysMonitor AMQP binding exists in this implementation; do not infer one from the dashboard's SysMonitor view or invent an associated telemetry contract.

2 References and qualification

See `docs/source-manifest.json` for original/relocated source SHA256 records, `VERIFICATION.md` for checks actually run, and `docs/LocalCloudManager.sysml` for the descriptive, unparsed variability model. No Java interchangeability, native TLS or sysMonitor publisher is claimed. Existing cloud source and running deployment remain unchanged.